

MAZONG SHAN, China

WMO: 523230

Lat: 41.8048N

Lon: 97.0320E

Elev: 1770

StdP: 81.79

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.0	-21.7	-32.3	0.2	-16.3	-29.9	0.3	-15.2	13.2	-6.9	11.9	-7.3	3.4	270	0.668

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	30.8	13.4	29.4	12.7	28.0	12.2	15.3	24.3	14.4	23.8	13.7	23.6	5.1	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
12.6	11.3	17.5	11.3	10.3	17.5	9.8	9.4	17.3	49.3	24.4	46.4	23.7	44.2	23.5	22.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
12.3	10.6	9.3	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-28.3	33.3	3.2	1.3	-30.6	34.2	-32.4	34.9	-34.2	35.7	-36.5	36.6
			-28.6	16.8	3.1	1.8	-30.8	18.1	-32.6	19.2	-34.4	20.2	-36.6	21.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.6	-11.4	-7.4	-1.0	6.8	13.0	18.7	21.4	19.6	13.3	5.2	-2.7	-9.4	(6)
(7)		DBStd	12.05	4.62	4.54	5.21	4.83	3.98	3.02	2.73	2.93	3.72	4.31	4.82	4.87	(7)
(8)		HDD10.0	2774	662	488	341	117	18	1	0	0	12	156	380	601	(8)
(9)		HDD18.3	4864	921	721	599	347	168	31	6	21	156	407	630	859	(9)
(10)		CDD10.0	1162	0	0	0	19	112	262	353	298	110	7	0	0	(10)
(11)		CDD18.3	208	0	0	0	0	3	42	100	59	3	0	0	0	(11)
(12)		CDH23.3	2314	0	0	0	5	86	487	1026	644	67	1	0	0	(12)
(13)		CDH26.7	614	0	0	0	0	8	102	339	158	7	0	0	0	(13)
(14)	Wind	WSAvg	4.4	4.2	4.1	4.6	5.1	4.9	4.5	4.2	4.1	4.0	4.2	4.7	4.6	(14)
(15)	Precipitation	PrecAvg	76	1	1	2	3	6	16	24	15	5	2	1	1	(15)
(16)		PrecMax	157	4	5	15	12	22	54	98	30	18	15	6	4	(16)
(17)		PrecMin	36	0	0	0	0	0	1	3	0	0	0	0	0	(17)
(18)		PrecStd	29	1	1	3	4	6	12	20	9	5	3	2	1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.9	8.7	16.4	23.7	27.8	30.9	33.4	31.8	27.6	21.2	13.5	5.2	(19)
(20)		MCWB	-2.0	-0.1	4.1	8.2	9.6	13.0	14.4	14.1	11.8	7.7	3.2	-1.8	(20)	
(21)		2%	DB	1.4	5.5	13.4	20.6	25.7	29.2	31.7	29.8	25.1	18.4	10.7	2.7	(21)
(22)		MCWB	-4.1	-1.7	2.8	6.7	9.4	11.9	13.8	12.9	10.1	6.4	1.8	-3.1	(22)	
(23)		5%	DB	-1.3	3.3	11.0	18.3	23.8	27.7	30.1	28.5	23.4	16.2	8.3	0.3	(23)
(24)		MCWB	-5.8	-2.9	1.6	5.8	8.7	11.6	13.3	12.4	9.4	5.4	0.6	-4.5	(24)	
(25)		10%	DB	-3.8	1.1	8.6	16.1	21.9	26.4	28.6	27.0	21.6	14.0	5.7	-2.0	(25)
(26)		MCWB	-7.3	-4.2	0.3	4.8	8.2	11.1	12.8	12.1	8.8	4.2	-0.7	-5.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.5	0.1	4.7	9.0	11.7	15.6	17.6	16.3	13.4	8.2	3.6	-1.4	(27)
(28)		MCDWB	4.3	7.8	14.6	21.6	22.7	23.3	28.7	26.2	22.5	19.0	12.5	4.7	(28)	
(29)		2%	WB	-3.9	-1.5	3.1	7.4	10.3	14.2	15.7	14.9	11.6	6.9	2.0	-3.0	(29)
(30)		MCDWB	1.2	5.0	12.5	19.1	22.7	22.9	24.9	23.2	21.4	17.3	10.1	2.4	(30)	
(31)		5%	WB	-5.7	-2.7	1.8	6.2	9.4	13.2	14.7	13.9	10.5	5.7	0.7	-4.5	(31)
(32)		MCDWB	-1.5	2.9	10.4	16.9	22.0	22.7	24.4	24.0	20.8	15.2	7.9	0.0	(32)	
(33)		10%	WB	-7.3	-4.0	0.5	5.2	8.6	12.4	14.0	13.1	9.5	4.5	-0.6	-5.9	(33)
(34)		MCDWB	-3.8	0.9	8.2	15.0	20.6	22.7	23.8	23.9	20.0	13.2	5.5	-2.1	(34)	
(35)	Mean Daily Temperature Range		MDBR	13.3	14.3	14.6	13.6	13.2	12.2	11.8	12.2	13.4	14.1	12.9	12.6	(35)
(36)		5% DB	MCDBR	15.7	16.7	17.2	16.6	16.1	14.5	14.1	14.4	15.6	16.7	15.9	14.7	(36)
(37)		MCWBR	11.2	11.1	9.9	8.4	7.4	5.6	4.7	5.3	6.5	8.5	9.2	10.2	(37)	
(38)		5% WB	MCDBR	15.3	15.9	15.9	14.9	13.7	11.2	11.2	11.6	13.1	15.3	15.4	14.2	(38)
(39)		MCWBR	11.1	10.6	9.1	7.5	6.6	4.6	4.0	4.6	5.9	8.0	9.1	9.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.273	0.317	0.367	0.415	0.422	0.437	0.430	0.406	0.379	0.332	0.297	0.271	(40)	
(41)		taud	2.295	2.177	2.034	1.947	1.993	2.024	2.089	2.139	2.158	2.221	2.267	2.301	(41)	
(42)		Ebn at Noon	875	877	868	851	855	840	842	851	847	848	831	844	(42)	
(43)		Edh at Noon	87	116	151	178	175	170	158	145	132	109	89	79	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.53	3.61	5.07	6.23	7.17	7.09	6.83	6.30	5.43	4.20	2.80	2.22	(44)	
(45)		RadStd	0.13	0.16	0.22	0.37	0.34	0.46	0.36	0.24	0.26	0.12	0.14	0.12	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+66	N/A

Nomenclature: See separate page